

Data Visualization

INTELLIGENT SYSTEMS

ESCUELA POLITÉCNICA DE INGENIERÍA DE GIJÓN



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Introduction

Basic Concepts

Types of Visualization

Guidelines

Summary

References

Introduction

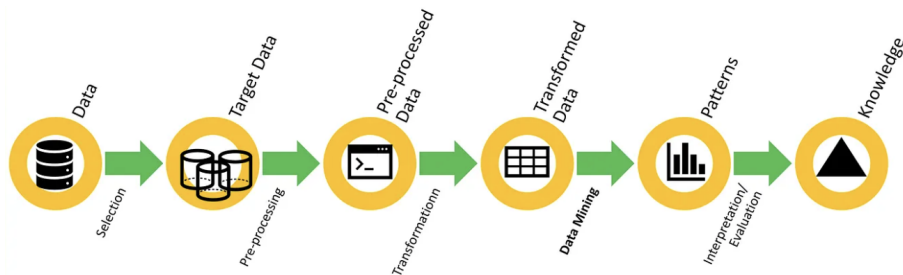
Visualization

- Converting abstract information into images
- Main goals:
 - * Summarize large amounts of data
 - * Extract new insights and patterns
 - * Make information easier to understand and communicate
- Preprocessing typically includes:
 - * Better generalization
 - * Faster learning
 - * More robust models

Knowledge Discovery in Databases (KDD)

→ KDD defines the stages required to extract knowledge from data

→ Main phases:



Motivating Example

→ Google search popularity for “vacations”

* As a table, it is hard to detect trends, seasonality, or peaks

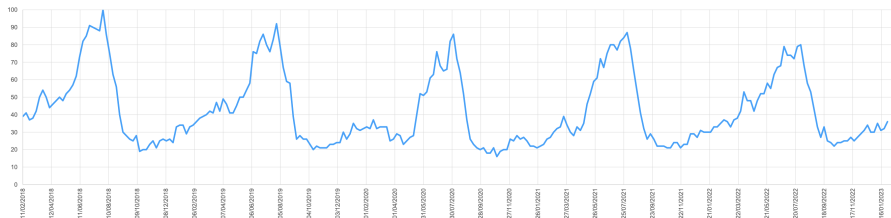
Week	9	10	11	12	...	53	...	9	...
Popularity	37	38	41	49	...	44	...	39	...

(Excerpt from a longer time series)

Motivating Example

→ A plot makes patterns visible:

- * Peaks, trends, and seasonality
- * Outliers and sudden changes



Motivating Example

→ Iris measurements (raw data)

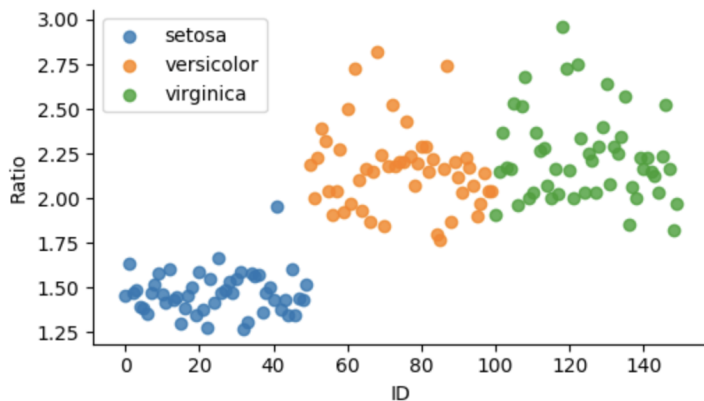
* As a table, structure is not obvious

sepal_length	sepal_width	petal_length	petal_width	class
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	virginica
4.7	3.2	1.3	0.2	versicolor
4.6	3.1	1.5	0.2	setosa
⋮	⋮	⋮	⋮	⋮

Introduction

Motivating Example

→ A scatter plot reveals class structure



Basic Concepts

Types of Variables

→ Example dataset:

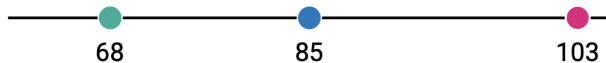
Weight	Height	Children	Hair color
85	181	3	brown
68	173	0	blond
103	155	1	brown

→ Numerical variables: continuous or discrete

→ Categorical variables

Representing One Variable

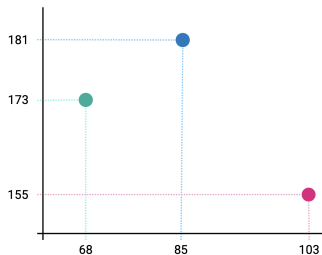
- Single numerical variable: Weight
- Goal: understand its distribution
- Typical visualization: Histogram / Bar chart



Representing Two Variables

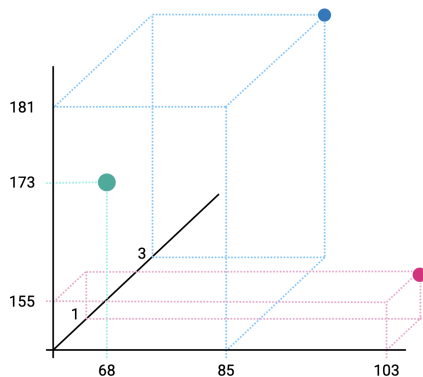
→ Relationship between two numerical variables: Weight vs Height

→ Typical visualization: Scatter plot



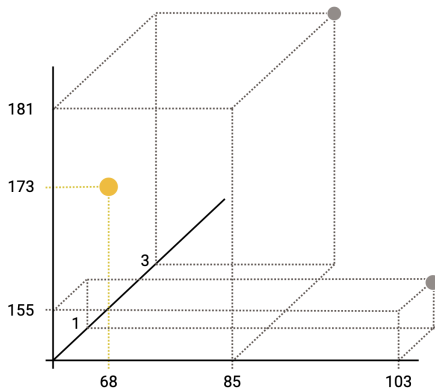
Representing Three Variables

- Third variable (Children) can be encoded using: color, size, marker shape
- Typical visualization: Scatter plot + color / size encoding



Representing Four Variables

- Multiple visual channels are combined: color, size, shape, position (x, y)
- Complexity increases quickly: Multi-encoded Scatter plot



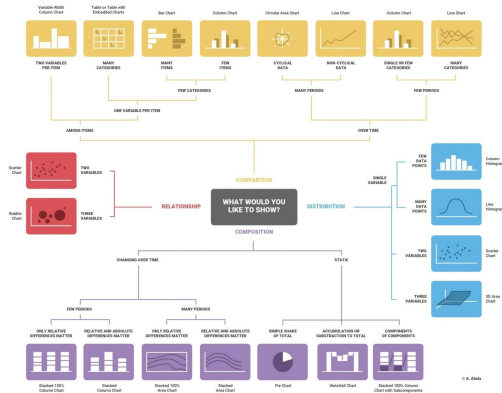
Types of Visualization

Types of Visualization

Types of Visualization

→ The choice of visualization depends on:

- * The type of variables
- * The relationship we want to show
- * The message we want to communicate



Types of Visualization

Types of Visualization: Distribution

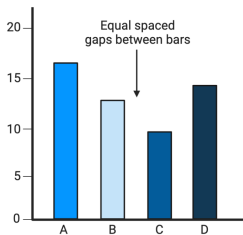
→ Show the possible values of a variable and their frequency

→ Typical visualizations:

- * Bar chart (categorical variables)
- * Histogram (numerical variables)

Bar Chart

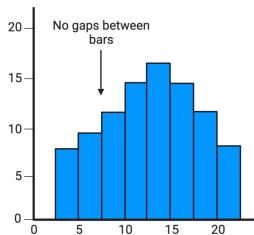
Shows comparison of 2+ categories



X-axis: categories
Y-axis: numerical values

Histogram

Shows frequency distribution of data

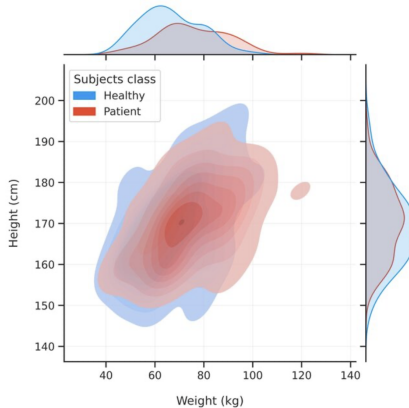


X-axis: intervals of continuous number
Y-axis: numerical values

Types of Visualization

Types of Visualization: Distribution (Joint Density)

- Show the joint distribution of two variables
- Useful to identify dense regions and correlations
- KDE for two numerical variables:



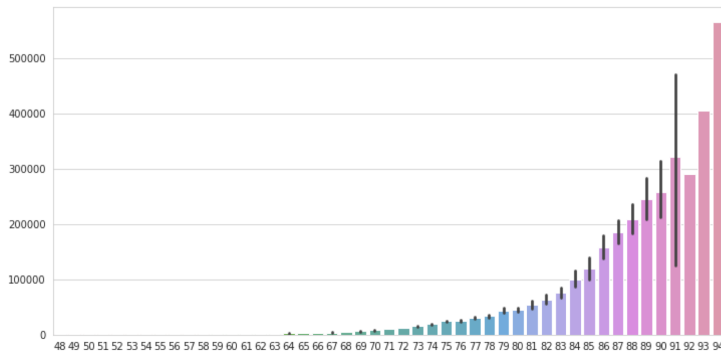
Types of Visualization

Types of Visualization: Relationship or Dependency

→ Represent the relationship between variables (typically two)

→ Common visualizations:

- * Bar chart:



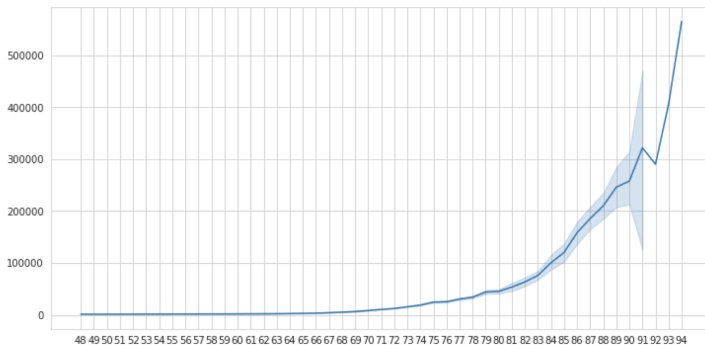
Player rating *vs* Salary

Types of Visualization

Types of Visualization: Relationship or Dependency

→ Common visualizations:

* Line chart:



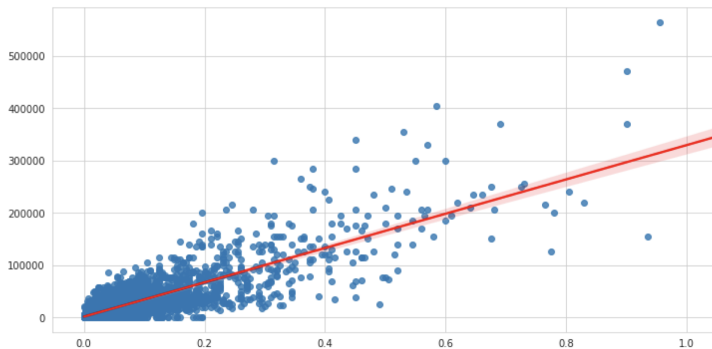
Player rating vs Salary

Types of Visualization

Types of Visualization: Relationship or Dependency

→ Common visualizations:

* Scatter plot:



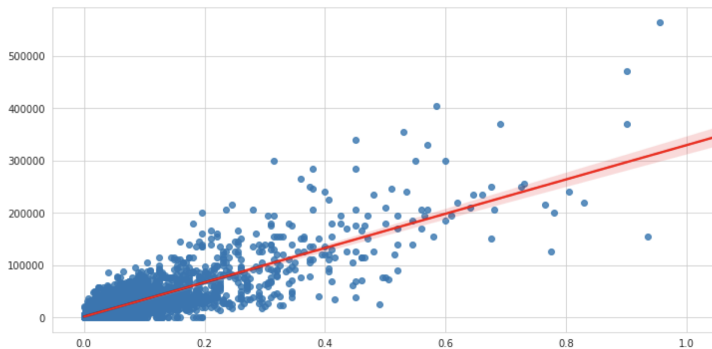
Market value *vs* Salary

Types of Visualization

Types of Visualization: Relationship or Dependency

→ Common visualizations:

* Scatter plot:



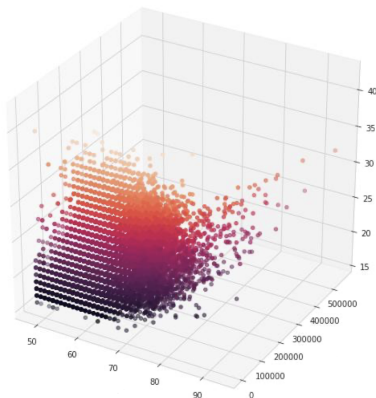
Market value *vs* Salary

Types of Visualization

Types of Visualization: Relationship or Dependency

→ Common visualizations:

- * 3D-Scatter plot:



Market value *vs* Salary *vs* Age

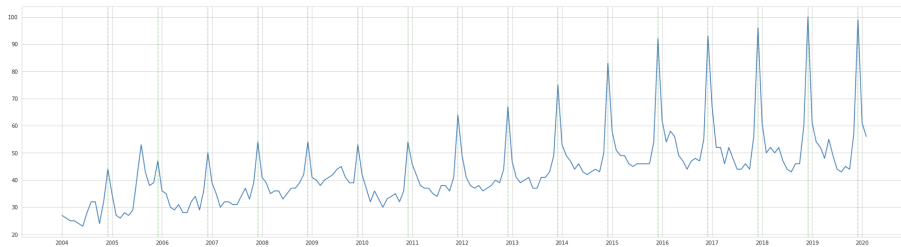
Types of Visualization

Types of Visualization: Trend

→ Show the evolution of a variable over time

→ Typical visualizations:

- * Line chart



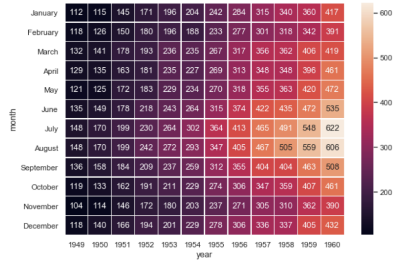
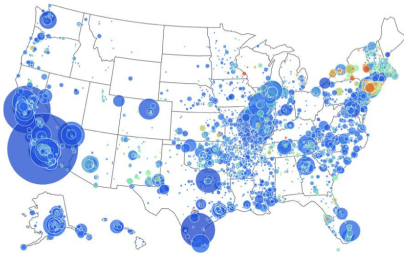
Google search popularity *vs* Time

Types of Visualization

Types of Visualization: Trend

→ Other visualizations:

- * Geographical visualizations (maps)
- * Heatmaps
- * Network graphs

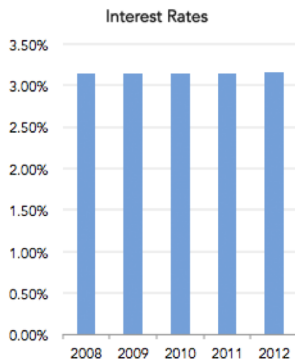


Guidelines

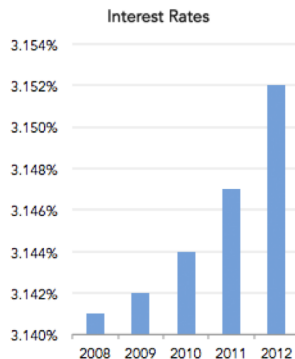
Guidelines

Guidelines: Axes

- Do not manipulate axes to your advantage
- * Always use an appropriate scale on the vertical axis
 - * Truncated axes can exaggerate or hide differences



Yes



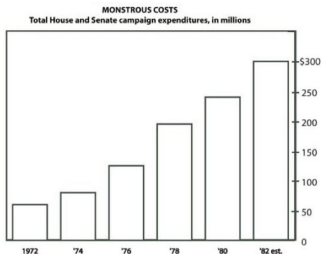
No

Guidelines

Guidelines: Avoid Chart Junk

→ Avoid excessive decoration

- * Use traditional, clean charts that do not distract
- * Decorative elements should not compete with the data



Yes



No

Guidelines: Comparing Values

- Choose the right chart for comparison
 - * Use bar charts to compare values or changes over time
 - * Avoid pie charts when precise comparison is required

Guidelines: Proportions

- Show proportions clearly
 - * Use stacked area charts to show proportions over time
 - * Avoid individual area charts for direct comparison

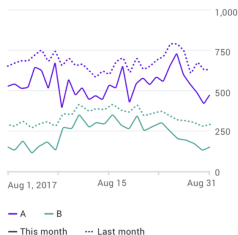
Guidelines

Guidelines: Consistent use of Color and Style

→ Use color and style to guide interpretation

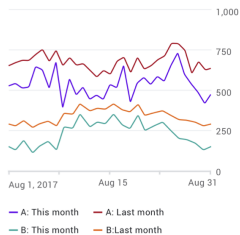
- * Use color to highlight relevant areas or important trends
- * Avoid using too many colors at the same time
- * Use the same color to represent the same element
- * Use different styles (line type, markers, patterns) to compare elements
- * Do not change colors arbitrarily across visualizations

All Campaigns



Yes

All Campaigns



No

Summary

Summary

- Data visualization allows us to:
 - * Understand how data behaves
 - * Identify patterns, trends, and outliers
 - * Extract knowledge without building learning models
- To visualize data effectively, it is necessary to:
 - * Know the type of variables involved
 - * Select the appropriate type of visualization
 - * Clearly describe what is being shown (titles, axes, legends)
- Good practices:
 - * Do not manipulate axes
 - * Avoid visual overload
 - * Use color and visual encodings consistently

References

- [1] BISHOP, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006.
- [2] FEW, S. *Show Me the Numbers: Designing Tables and Graphs to Enlighten*. Analytics Press, 2012.
- [3] GÉRON, A. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems*. O'Reilly Media, Inc., 2022.
- [4] HASTIE, T., TIBSHIRANI, R., AND FRIEDMAN, J. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, 2009.
- [5] RASchKA, S., AND MIRJALILI, V. *Python machine learning: Machine learning and deep learning with Python, scikit-learn, and TensorFlow 2*. Packt publishing ltd, 2019.
- [6] TUFTE, E. R. *The Visual Display of Quantitative Information*. Graphics Press, 2001.
- [7] VANDERPLAS, J. T. *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, Inc., 2016.
- [8] WILKINSON, L. *The Grammar of Graphics*. Springer, 2005.